
Quantitative Finance From Foundations to Research

A mathematical and empirical laboratory for financial engineering

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Version: 1.0, 2026

Theory is a model; evidence is a discipline.

How to use this book

This text is a study companion, not investment advice. Every model is conditional on assumptions, calibration choices, and data quality. Read each theorem with its hypotheses, reproduce each numerical result, and treat out-of-sample validation as part of the mathematics.

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Chapter 1

Mathematical foundations

1.1 Probability and information

A probability space is $(\Omega, \mathcal{F}, \mathbb{P})$, where events in $\mathcal{F}$ receive probabilities. A filtration $(\mathcal{F}_t)_{t \geq 0}$ represents information available through time. A random variable X is measurable with respect to $\mathcal{F}_t$ when its value is known at t .

Definition 1.1 (Conditional expectation). *For integrable X , $\mathbb{E}[X \mid \mathcal{G}]$ is the $\mathcal{G}$ -measurable random variable satisfying $\mathbb{E}[\mathbf{1}_A \mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[\mathbf{1}_A X]$ for every $A \in \mathcal{G}$.*

Theorem 1.2 (Tower property). *If $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$ and X is integrable, then $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}]$ almost surely.*

1.2 Linear algebra and optimization

For a portfolio $w \in \mathbb{R}^n$ and covariance matrix $\Sigma \succeq 0$, variance is $w^\top \Sigma w$. The constrained minimum-variance problem is

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad \mu^\top w = \mu_0.$$

The Lagrangian and first-order conditions are sufficient when the objective is convex and constraints are affine. With only full investment, $w^* = \Sigma^{-1} \mathbf{1} / (\mathbf{1}^\top \Sigma^{-1} \mathbf{1})$ when Σ is positive definite.

1.3 Calculus and differential equations

Assumption 1.3 (Regular diffusion). *Let S_t satisfy $dS_t = \mu(t, S_t) dt + \sigma(t, S_t) dW_t$, where coefficients are measurable, Lipschitz in state, and have linear growth.*

Theorem 1.4 (Ito's formula). *For $f \in C^{1,2}$, $df(t, S_t) = (f_t + \mu f_s + \frac{1}{2} \sigma^2 f_{ss}) dt + \sigma f_s dW_t$.*

The second derivative term is the distinctive feature of stochastic calculus. Applying Ito to $\log S_t$ gives $d \log S_t = (\mu - \frac{1}{2} \sigma^2) dt + \sigma dW_t$.

1.4 No-arbitrage and change of measure

Definition 1.5 (Equivalent martingale measure). *A probability measure $\mathbb{Q}$ is equivalent to $\mathbb{P}$ if they have the same null events. It is risk-neutral for a discounted asset $\tilde{S}_t$ when $\tilde{S}_t$ is a $\mathbb{Q}$ -martingale.*

Theorem 1.6 (Risk-neutral valuation). *In a complete arbitrage-free market, the time- t value of a payoff H at T is $V_t = \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^T r_u du} H \mid \mathcal{F}_t]$.*

1.5 Black–Scholes as a benchmark

For constant r, σ and no dividends, a European call $C(S, t)$ solves

$$C_t + rSC_s + \frac{1}{2}\sigma^2 S^2 C_{ss} - rC = 0, \quad C(S, T) = (S - K)^+.$$

The closed form is $C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$, with $d_{1,2} = [\log(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau]/(\sigma\sqrt{\tau})$. Its assumptions exclude jumps, stochastic volatility, liquidity costs, and model uncertainty; the formula is a benchmark, not a law of nature.

Chapter 2

Macroeconomics for markets

2.1 Intertemporal choice

A representative household maximizes $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$ subject to $a_{t+1} = (1 + r_t)a_t + y_t - c_t$. The Euler equation is $u'(c_t) = \beta(1 + r_{t+1}) \mathbb{E}_t[u'(c_{t+1})]$. For CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$, expected consumption growth is tied to the real short rate and risk aversion.

2.2 IS–Phillips–policy system

A compact linearized system is $x_t = \mathbb{E}_t x_{t+1} - \sigma(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n)$ and $\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t$, with policy $i_t = \rho i_{t-1} + (1 - \rho)(\phi_\pi \pi_t + \phi_x x_t) + \varepsilon_t$. The symbols encode the output gap, inflation, nominal rate, natural real rate, and shocks.

Remark 2.1 (Identification). *A correlation between rates and returns is not automatically a monetary-policy causal effect. Expectations, simultaneous responses, and measurement error must be modeled.*

2.3 Term structure

Under no arbitrage, a zero-coupon bond price is $P(t, T) = \mathbb{E}_t^{\mathbb{Q}}[e^{-\int_t^T r_s ds}]$. Its continuously compounded yield is $y(t, T) = -\log P(t, T)/(T - t)$. Duration approximates price sensitivity: $\Delta P/P \approx -D \Delta y + \frac{1}{2}C(\Delta y)^2$.

2.4 State-space thinking

Latent economic states can be estimated with $x_{t+1} = Ax_t + \eta_{t+1}$ and $y_t = Cx_t + \varepsilon_t$. The Kalman filter is optimal for linear Gaussian systems; otherwise particle or nonlinear filters are needed. Always distinguish a filtered estimate, which uses information through t , from a smoothed estimate, which uses future observations.

Chapter 3

Microeconometrics and empirical finance

3.1 Potential outcomes and causal estimands

For unit i , let $Y_i(1), Y_i(0)$ be potential outcomes. The average treatment effect is $\tau = \mathbb{E}[Y(1) - Y(0)]$. Observed data reveal only $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$, so causal identification requires assumptions.

Assumption 3.1 (Unconfoundedness). $(Y(1), Y(0)) \perp D \mid X$, with overlap $0 < \mathbb{P}(D = 1 \mid X = x) < 1$ on the support.

Under this assumption, regression adjustment or inverse-probability weighting can identify τ . Financial examples include evaluating a policy announcement or a portfolio signal, but time dependence and interference require care.

3.2 Time series and returns

The AR(1) model $r_t = c + \phi r_{t-1} + \varepsilon_t$ is covariance stationary when $|\phi| < 1$. For volatility, GARCH(1,1) specifies $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, with $\omega > 0$, $\alpha, \beta \geq 0$, and $\alpha + \beta < 1$ for finite unconditional variance.

3.3 Regression discipline

For $y = X\beta + \varepsilon$, OLS is unbiased conditional on X if $\mathbb{E}[\varepsilon \mid X] = 0$. Heteroskedasticity and autocorrelation invalidate naive standard errors; use robust or HAC estimators and report effective sample size. Multiple testing, data snooping, survivorship bias, and look-ahead bias are first-order threats in strategy research.

Theorem 3.2 (Law of large numbers, informal). *For stationary ergodic observations with finite first moment, sample averages converge almost surely to population expectations. Dependence does not disappear by pretending observations are iid; the assumptions determine the limit theory.*

3.4 Validation

Use expanding or rolling walk-forward splits. A signal is credible only after transaction costs, turnover, liquidity, missingness, and parameter instability are included. Bootstrap procedures should

preserve relevant dependence, for example via moving blocks rather than independently resampling daily returns.

Chapter 4

Financial markets and engineering practice

4.1 Market instruments

Equities are claims on residual cash flows; bonds promise contractual cash flows; forwards and futures transfer exposure to a future price; options create nonlinear payoffs. A payoff diagram is often the fastest way to expose hidden leverage and tail risk.

4.2 Risk measures

For loss L , Value at Risk at level α is $\text{VaR}_\alpha(L) = \inf\{\ell : \mathbb{P}(L \leq \ell) \geq \alpha\}$. Expected Shortfall is $\text{ES}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)]$ when the conditional expectation is well-defined. Historical, parametric, and Monte Carlo estimates answer different questions.

Proposition 4.1 (Subadditivity of Expected Shortfall). *Under standard integrability conditions, $\text{ES}_\alpha(X + Y) \leq \text{ES}_\alpha(X) + \text{ES}_\alpha(Y)$. This makes Expected Shortfall a coherent risk measure alongside monotonicity, positive homogeneity, and translation invariance.*

4.3 Portfolio construction

Mean-variance optimization is $\max_w w^\top \mu - \frac{\lambda}{2} w^\top \Sigma w$ subject to constraints. Estimation error can dominate the theoretical solution, so shrinkage covariance estimates, robust constraints, and out-of-sample tests are essential. Risk parity targets contributions $RC_i = w_i(\Sigma w)_i / (w^\top \Sigma w)$ rather than raw weights.

4.4 Numerical methods

A Monte Carlo estimator for $V = e^{-rT} \mathbb{E}[h(S_T)]$ is $\hat{V} = e^{-rT} n^{-1} \sum_{j=1}^n h(S_T^{(j)})$ with standard error $s/\sqrt{n}$. Antithetic variates, control variates, stratification, and quasi-random sequences reduce variance but do not cure a misspecified model.

4.5 Calibration and model risk

Calibration minimizes a weighted discrepancy between observed and model prices, often $\min_{\theta} \sum_i w_i (P_i^{\text{model}}(\theta) - P_i^{\text{mkt}})^2$. A low calibration error is not validation: examine residual structure, stability across dates, and hedging performance. Greeks such as $\Delta = \partial V / \partial S$ and $\Gamma = \partial^2 V / \partial S^2$ quantify local sensitivity, while stress tests explore nonlocal moves.

4.6 Market microstructure

Observed prices reflect bid–ask spreads, discrete trading, latency, price impact, and constraints. A backtest must execute at plausible prices and align signal timestamps with the next available trade. A strategy with positive frictionless alpha can have negative net performance after costs.

Research checklist

Before trusting a result: state the estimand; identify the filtration and information set; write the assumptions; separate training, validation, and test periods; report uncertainty; stress the model; test sensitivity to costs, missing data, and dependence; and document provenance.